Example 6.29 As observed from the top of a 60 m high lighthouse from the sea level, the angles of depression of two ships are $28°$ and $45°$. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships. ($\tan 28° = 0.5317$)

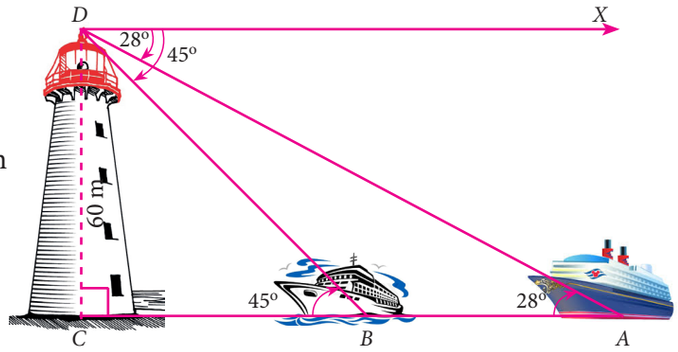


Fig. 6.24

Solution Let the observer on the lighthouse CD be at D.

Height of the lighthouse $CD = 60$ m

From the diagram,

$\angle XDA = 28° = \angle DAC$ and

$\angle XDB = 45° = \angle DBC$

In the right angled ΔDCB, $\tan 45° = \dfrac{DC}{BC}$

$$1 = \frac{60}{BC} \Rightarrow BC = 60 \text{ m}$$

In the right angled ΔDCA, $\tan 28° = \dfrac{DC}{AC}$

$$0.5317 = \frac{60}{AC} \Rightarrow AC = \frac{60}{0.5317} = 112.85$$

Distance between the two ships $AB = AC - BC = 52.85$ m

Example 6.30 A man is watching a boat speeding away from the top of a tower. The boat makes an angle of depression of $60°$ with the man's eye when at a distance of 200 m from the tower. After 10 seconds, the angle of depression becomes $45°$. What is the approximate speed of the boat (in km / hr), assuming that it is sailing in still water? ($\sqrt{3} = 1.732$)

Solution Let AB be the tower.

Let C and D be the positions of the boat.

From the diagram,

$\angle XAC = 60° = \angle ACB$ and

$\angle XAD = 45° = \angle ADB$, $BC = 200$ m

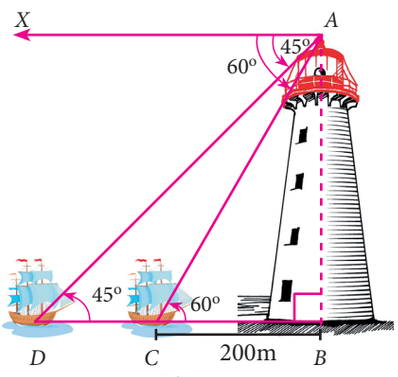


Fig. 6.25

In the right angled ΔABC, $\tan 60° = \dfrac{AB}{BC}$

$$\Rightarrow \quad \sqrt{3} = \frac{AB}{200}$$

we get $AB = 200\sqrt{3}$ ...(1)

In the right angled ΔABD, $\tan 45° = \dfrac{AB}{BD}$

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